Supplementary material: Text properties and children's reading pace in a multilingual storytelling app

Table of contents

S1. The theory object 1

S2. Studies from the accompanying systematic review 2

S3. Language resources for the text features 3

S4. The literature sizing 5

S5. The design analysis in full 5

S6. Deriving the priors from the systematic review 7

S7. Data-access certification 8

References 8

S1. The theory object

The constructs, causal claims, auxiliary assumptions, rival accounts and predictions behind this protocol are held in a single machine-readable file, `theory/intract.theory.yaml`, which the theoryforge package reads and checks[1]. Figure 2 of the main text is emitted from that file, so the diagram cannot disagree with the hypotheses. Running `theory/build_theory.R` reproduces everything in this section.

The object registers twelve constructs, each with a definition, its intended measurement and its boundary conditions. Eighteen propositions link them, each carrying the mechanism it claims and not only a direction. Fifteen of the eighteen are directed edges of the causal graph, and the other three are moderation claims. Six auxiliary assumptions are recorded as entries in their own right, five of them corresponding to questions put to the data controller and listed in Table 2 of the main text. Six rival accounts are registered, and seven of the ten predictions are marked diagnostic against one of them.

The rigour checklist scores the object at 90.9 out of 100 with no blocking failure, and every item passes. On falsifiability and precision, at least one prediction forbids an observation, all ten predictions are interval predictions, and their mean severity clears the threshold. On construct quality, every construct carries a definition, a measurement and boundary conditions, no pair exceeds the lexical redundancy screen (largest observed similarity 0.14), and the theory's scope is stated. On structure, every proposition states a mechanism, at least one relation is causal, auxiliary assumptions are fewer than propositions with none added defensively

in response to a prediction, and every prediction is reachable from a proposition in the graph. Finally, at least one prediction discriminates between registered alternatives, and a formal-model reference points at the design specification used for the precision analysis.

Because the causal graph is acyclic it yields testable implications, thirty of them, of which three are tested in the fourth outcome-neutral quality check of the main text. The most consequential is that the three text properties should be mutually independent once the reading level and the generation route are conditioned on. A first version of the object, lacking the text-provenance construct, implied that they were independent unconditionally, which is false of any real corpus: a text's target reading level shifts all three property families at once. Adding text provenance as a common cause of all three, and giving it an incoming path from the reader's own decoding automaticity, both repaired that implication and derived the adjustment set the analysis uses.

S2. Studies from the accompanying systematic review

The Introduction summarises the evidence that motivates the feature set and the moderators. Table 1 lists the studies closest to this design in method, meaning a broad set of text features entered into a statistical or machine-learning model to predict a behavioural or comprehension outcome in children.

Table 1

Supplementary Table 1. Studies modelling children's reading computationally from text features

Study	Sample and design	Method and key findings
[2]	Children (*N* = 158, ages 8;8 to 11;2); multilevel logistic models	Word frequency, concreteness, deep cohesion and lexical arousal each predicted passage comprehension after decoding and language comprehension were controlled.
[3]	Children (*N* = 1,229, grades 2 to 9); multivariate regression	Text and sentence characteristics predicted comprehension, but the direction and size of each effect varied by question type.
[4]	Children (*N* = 924, grade 4); non-linear regression	Estimated lexical text coverage predicted comprehension non-linearly, rising more steeply above roughly 56 per cent coverage.
[5]	Elementary school students; feature-based prediction model	Sentence complexity, lexical surprisal and rare-word features improved prediction of item-level comprehension scores over a baseline.
[6]	Bilingual children (*N* = 44, 22 with language impairment); eye-tracking	Syntactic complexity impaired comprehension accuracy and slowed recall, disproportionately for children with language impairment.

Table 2 lists the digital and computerised reading studies closest to this design in modality and measurement. Most come from the accompanying systematic review, with one directly comparable app-log precedent retained from outside it[7].

Table 2

Supplementary Table 2. Digital and app-based reading studies informing the present design

Study	Platform and sample	Data, focus and key findings
[7]	Educational app (BookBites); children $N \approx 48{,}000$, grades 3 to 6	App reading time against standardised test scores. Reading speed correlated with ability ($r = .39$ to $.48$); models predicted 16 to 18 per cent of variance in progress.
Gissel and Andersen[8]	Digital text-to-speech tool; $N = 1{,}013$ children, 47 classrooms, grade 2	Cluster-randomised trial against an active phonics control. No difference in decoding, but the digital tool improved comprehension, which is the outcome most sensitive to text-level scaffolding.
[9]	Computerised book-reading assessment; $N = 852{,}295$ students, 3,243 schools	Quiz-based comprehension checks linked to a computerised adaptive test. Higher quiz engagement went with higher achievement, but the engagement measure itself was of limited reliability.
[10]	Online reading assessment; multi-country sample of fourth graders (ePIRLS 2016)	Explanatory item response modelling of item format, text complexity and cognitive complexity. Item properties and student characteristics interacted to predict online comprehension.
[11]	Text-to-speech study; children aged 8 to 12 with reading and language difficulties	Within-subject comparison of silent reading, listening only and text-to-speech. Text-to-speech improved comprehension over silent reading, with the benefit concentrated in children with dyslexia.
Tüchler and Cain[12]	Digital against print medium; $N = 120$ children, grades 4 to 6	Generalised linear mixed models of medium and text structure. Comprehension was lower in the digital condition after word recognition and vocabulary were controlled, a screen-inferiority effect relevant to interpreting any digital-only dataset.
Beigman Klebanov and Hoang[13]	App-embedded stealth assessment (Relay Reader)	Machine-learning models using text and oral-reading features predicted passage comprehension with over 80 per cent precision, showing that unobtrusive in-app comprehension prediction is feasible.

S3. Language resources for the text features

A multilingual design faces a measurement problem that a monolingual one does not, and it is not solved anywhere in the literature. Concreteness taken from a 40,000-item English rating study, a 350,000-lemma German regression, a 1,400-item Spanish rating study and a 77-language embedding prediction are four different measurements wearing one label. If a resource is chosen language by language on the grounds of local quality, then the method of measurement is perfectly confounded with the language, and standardising within language removes the difference in location and scale but not the difference in construct. There are two clean responses: use one uniformly produced resource everywhere, accepting lower absolute quality for higher comparability, or use native norms and treat every cross-language comparison as descriptive. This protocol takes the first, because language enters the confirmatory models as a grouping factor with varying slopes, and those slopes have to mean the same thing in every language. The native-norm sensitivity analysis and the per-language correlation between the two measures are specified in the main text. That correlation is the quantity that tells a reader how much the choice mattered.

Two consequences of that rule revise what an earlier version of this protocol assumed. The `wordfreq` package is not used as the primary frequency source: its data are a frozen snapshot of usage to about 2021, its maintainer discontinued it in September 2024 to avoid ingesting post-2022 web text contaminated by generated language, and its wordlists for Korean and Greek are built from fewer sources than the rest. The non-English editions of the NRC Valence, Arousal and Dominance lexicon are not used at all: they are machine translations of the English norms made in 2022, not ratings collected in the target languages, and their licence is non-commercial.

Table 3 gives the primary resource for each feature family, the sensitivity resource, and the coverage across the ten languages the structural sample showed. Where the export turns out to carry more languages, the same rules are applied to them and the table is extended to cover them.

Table 3

Supplementary Table 3. Resources for the text features, the uniform primary and the native sensitivity

Feature family	Primary, uniform across languages	Sensitivity, native where it exists	Coverage of the ten sampled languages
Word frequency	subs2vec, one OpenSubtitles pipeline applied identically to 55 languages[14]	The SUBTLEX family, native and peer-reviewed per language	All ten for the primary. Native norms exist for nine; Korean has none.
Age of acquisition	The cross-linguistic norms of Łuniewska and colleagues, an identical 299-word item set rated on an identical scale in 32 languages[15,16]	Broad native norm sets, which exist at scale for English, Dutch and Polish only	Nine of ten for the anchor set; Korean appears in neither paper. Because 299 words is thin for a text-level mean, age of acquisition is not an anchor for the lexical component and contributes only where a broad native set exists.
Surprisal	Goldfish, an identical 125-million-parameter architecture trained on byte-premium-scaled equal-sized data for each language[17]	mGPT, as a single multilingual model rather than one model per language	All ten at the 1 GB tier. This is the one feature for which a comparable multilingual resource exists by construction.
Word length, lexical density, lexical diversity	Computed from the text directly, so no norm is involved	Not applicable	All ten.
Concreteness	Native human ratings only; no value is predicted or projected	Not applicable, since the native norm is the only acceptable measure	Nine of ten, at between roughly 1,400 and 40,000 items[18]. Korean has no indexed equivalent, so concreteness is not computed for Korean and Korean does not contribute to any concreteness analysis.
Dependency distance and clause structure	Stanza, whose per-language labelled attachment scores are published and range from 85.5 to 95.2 across these ten[19]	UDPipe 2, whose current models track a later Universal Dependencies release and score higher on all ten	All ten in both. Both parsers are run, and the per-language correlation between the two dependency-distance estimates is reported with the feature-pipeline validation. A language in which they disagree materially is flagged and its grammatical component is not used.

Feature family	Primary, uniform across languages	Sensitivity, native where it exists	Coverage of the ten sampled languages
Adjacent-sentence similarity	BGE-M3, peer-reviewed, over 100 languages, and able to take a whole passage as input[20]	multilingual-E5-large-instruct, identified by checkpoint, since no DOI is registered	All ten in both. LaBSE is not used: it is optimised for retrieving translation pairs rather than for semantic similarity within a language.
Referential overlap, connective density	Computed from the parse and from language-specific connective inventories compiled for this study and deposited with it	Not applicable	All ten, with the inventory for each language checked by a native speaker before use.
Valence and arousal	MEmoLon, induced for 91 languages from a single procedure, at roughly two million entries per language[21]	Native rated norms, which exist for nine of the ten	All ten for the primary; Korean has no native norms, so only the primary is available there.

Korean is the weak point, and we record it here in advance. Of the seven norm-based feature families, Korean lacks a subtitle frequency norm, any coverage in the cross-linguistic age-of-acquisition set, any indexed concreteness or imageability norms, and any native valence and arousal norms. What it does have is a uniform frequency estimate, a Goldfish model, both parsers, induced affect and a lexical decision megastudy of 30,930 words. Korean therefore contributes to the lexical and grammatical components and to the affective predictors through the uniform resources, and contributes to no analysis that requires a native norm. A search of Crossref is a poor instrument for Korean-language psycholinguistics, so before the features are frozen a Korean-reading collaborator will search the Korea Citation Index and RISS directly, and any norm set found there will be added to the sensitivity analysis with the addition recorded as a deviation from this protocol. Greek is the next weakest: it has no native concreteness set for Standard Modern Greek, no CHILDES collection, and an affective lexicon of a different character from the rated norms available elsewhere.

Every tool is recorded by name, version and, where the tool is a model, by the exact checkpoint. Several of the tools named here have no registered DOI, which is normal for research software distributed through model repositories. Those are identified by checkpoint instead, and the identifiers are deposited with the feature-extraction code.

S4. The literature sizing

The claim in the Introduction that this study sits in an empty intersection is a count, and the queries behind it are reproduced verbatim in `literature/scopus_search_record.md`, together with the date of the search and the package version. Only count queries were issued, one request per cell, so nothing was downloaded and no record set was assembled. Figure 1 shows how the surrounding topics have moved over the window.

S5. The design analysis in full

The precision analysis simulated the planned crossed structure from the portable design specifications deposited in `design_analysis/`, and `design_analysis/run_design_analysis.R` reproduces it[22]. Full results for every grid point, including the Monte Carlo standard error and Wilson bounds on each decision probability and the counts of converged, singular and warning fits, are in

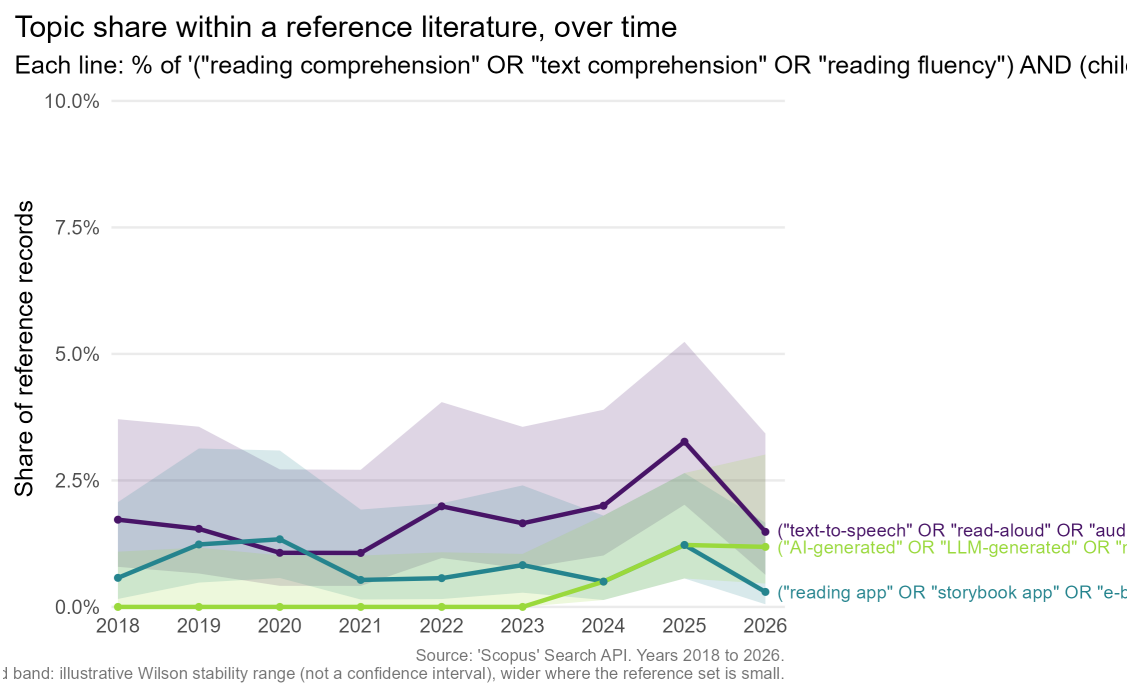


Figure 1: Supplementary Figure 1. Publication trend, 2018 to 2026, for children's reading comprehension in combination with narration, with reading apps and e-books, and with machine-generated text, as a percentage of the reference set. Machine-generated text does not appear in this literature at all before 2024.

`design_analysis/precision_sweep.csv`. Figure 2 shows the narration and within-child curves that the main text summarises.

The ordinal outcome behind Hypothesis 5 is handled by a separate script, `design_analysis/run_ordinal_analysis.R`, because a cumulative logit cannot be fitted by the routine the timing models use and treating a three-category rating as continuous would answer a different question. The script keeps the same design specification and the same seed discipline, generates the rating from it, retains each reading with the 30 per cent availability the structural sample showed, applies the protocol's five-rater floor inside every replicate, and fits the model the protocol specifies with `ordinal::clmm`.

The result is the most demanding requirement in the design. At 200, 400, 800 and 1,600 generated texts, with twelve readings each, the 95 per cent interval on a coefficient whose true value is 0.10 log odds has a mean width of 0.44, 0.30, 0.21 and 0.15, and the probability of a determinate verdict never rises above 0.29. The target of 0.90 is therefore not reached within the range simulated, and no crossing is quoted from it. What the grid does establish is the scaling: the fitted exponents of interval width on the number of texts are between -0.51 and -0.53 against a theoretical -0.5, and on that law the requirement projects to roughly 8,000 to 8,700 generated texts, which is about 2,300 to 2,500 texts carrying at least five ratings. Set against the 738 texts the same coefficient needs on reading time, the rating demands an order of magnitude more, which is what a three-category response observed for 30 per cent of readings costs. Both quantities depend on the rating's availability across the full window and on how heavily texts are reused, which the provider-computed counts will supply, and the simulation is re-run at the observed values before any outcome model is fitted. `design_analysis/summarise_ordinal.R` pools the cached grid points and reproduces every number in this paragraph.

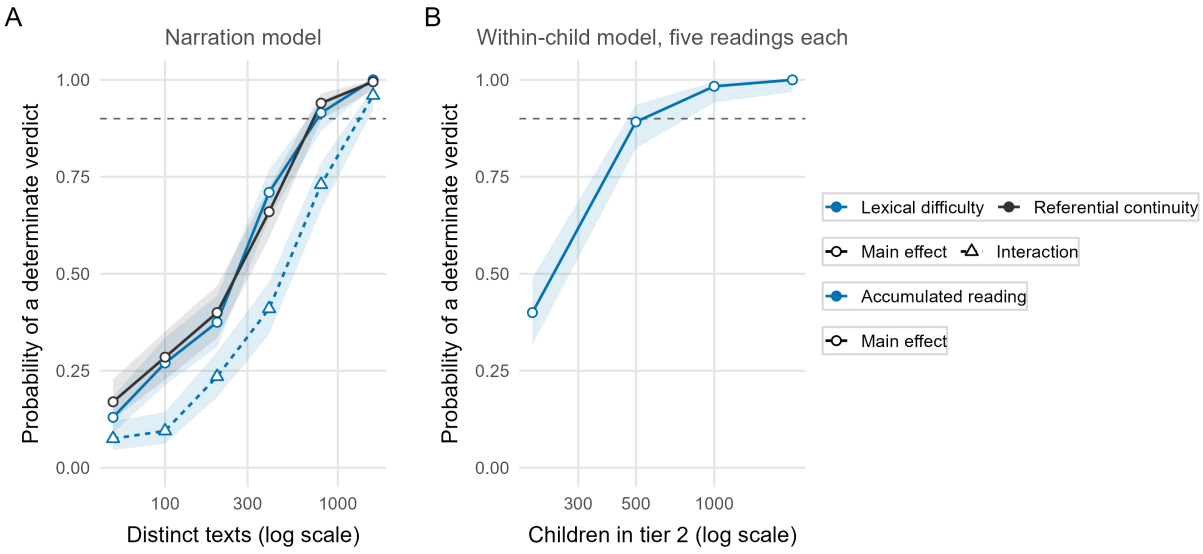


Figure 2: Supplementary Figure 2. Precision for the narration contrast (left) and the within-child accumulated-reading term (right). The narration curve is drawn for a balanced contrast and has to be read against the design effect for the observed narration prevalence, which is given in the main text.

S6. Deriving the priors from the systematic review

Priors for the confirmatory main effects come from the systematic review accompanying this project, through the pipeline whose freeze-and-run discipline is stated under Priors in the main text. This section records what the main text does not: the mapping from the review's extracted effects to this protocol's coefficients, the conversion order, and the pooling and de-duplication rules. The mapping is one-to-many and is fixed as follows. LexicalDifficulty pools the review's word-frequency, lexical-coverage, other-lexical and lexical-diversity effects on reading, sign-harmonised so that frequency and coverage are flipped to a difficulty direction while age of acquisition, word length and surprisal keep their sign. GrammaticalLoad pools syntactic-complexity and sentence-length effects. ReferentialContinuity pools coherence effects, with discourse-structure rows informing the secondary narrativity predictor instead. Valence and arousal pool the text-emotionality rows separately where the extraction distinguishes them.

Estimates are put on a common scale in a fixed order of preference: a standardised beta or a correlation goes through Fisher's *z*; a *d* or *g* converts to *r* using the effective sample size; an odds ratio converts to *d* and then to *r*. Where no effect size is reported but a test statistic is, *r* is recovered from *t* and its degrees of freedom and signed by the reported direction. Unstandardised coefficients without convertible scale information are not pooled and are listed in an exclusion appendix, as are rows whose direction the extraction could not resolve. Pooling is a random-effects meta-analysis within each combination of coefficient and outcome class, with one estimate per paper per relation, so that labelled subgroups of a single paper are never treated as independent. Where a meta-analysis in the corpus subsumes three or more of the corpus's own primary studies, the pooled row is preferred and the overlapping primaries dropped; otherwise the primaries are used and the pooled row dropped, and the alternative is reported as a sensitivity analysis.

The prior is then Student's $t(3, \hat{\mu}, \sigma)$ with $\hat{\mu}$ the pooled estimate on the model's scale and $\sigma = 2\sqrt{\mathrm{SE}(\hat{\mu})^2 + \hat{\tau}^2}$, floored at 0.10. The heavy tails let the data override a misplaced centre. The complete prior table, the forest plots, the conversion arithmetic and the list of excluded rows will accompany the Stage

2 report, and the pipeline's output is timestamped and deposited before the extract is transferred.

S7. Data-access certification

The text below was sent to the data controller on 7 August 2026 and is reproduced as it stands. The bracketed fields are completed and the document signed before this protocol is submitted. If any of them is still open at submission, we state that here.

> Gameloft confirms that, as of [date]:
>
> 1. An export of *Let's Story* reading sessions was built on [build date], covering [window] and comprising [n] rows, in the file or files listed in the accompanying record with their SHA-256 checksums.
> 2. That extract is held by Gameloft and has not been transferred, in whole or in part, to the University of Oxford research team or to anyone acting for it.
> 3. It will be transferred only on written confirmation of in-principle acceptance.
> 4. Material shared with the team to date: a 35-row sample and the data dictionary, shared on 27 July 2026 to confirm the structure of the table; the story texts; and the counts and file record described above.
> 5. No analysis relating story text to reading time has been conducted by Gameloft or reported to the team.
>
> Owner: [name, role]
> Deputy: [name, role]
> Signed: [name, role, date]

Two points arise from clause 4. The 35-row sample carries real reading times, and the main text discloses it directly. The story texts are named here because the editorial office confirmed in advance that stimulus materials may be received before submission, for the reasons given under Data provenance in the main text.

The accompanying record referred to in clause 1 is the manifest described in the main text, deposited with this protocol. Any row removed from the held extract between the build and the transfer is recorded by the data controller, as the Ethics statement sets out, and its effect on the counts will be reported at Stage 2.

References

1. Bernabeu, P. theoryforge: Systematic theory development. Zenodo https://doi.org/10.5281/zenodo.21229964 (2026).
2. Pickren, S. E., Stacy, M., Tufo, S. N. D., Spencer, M. & Cutting, L. E. The contribution of text characteristics to reading comprehension: Investigating the influence of text emotionality. *Read. Res. Q.* **57**, 649–667 (2022).
3. Aparicio, X., Mégalakaki, O., Drai-Zerbib, V. & Baccino, T. Comprehension performances of explanatory texts in French language according to their characteristics: Evidence for 1229 children from 2nd to 9th grade. *Sci. Stud. Read.* **26**, 287–304 (2022).
4. Ludewig, U., Hübner, N. & Schroeder, S. Vocabulary, text coverage, word frequency and the lexical threshold in elementary school reading comprehension. *Reading and Writing* **36**, 2409–2431 (2023).
5. Nie, Y., Deacon, H., Fyshe, A. & Epp, C. D. Predicting reading comprehension scores of elementary school students. https://doi.org/10.5281/ZENODO.6852952 (2022) doi:10.5281/ZENODO.6852952.

6. Wasito Kirana, A., Segoh, D., Achmad Syukri, M. & Prassetyo, S. Sentence processing in Indonesian-English bilingual children with and without language impairment. *GEMA Online® J. Lang. Stud.* **25**, 796–816 (2025).
7. Sortkær, B., Smith, E., Reimer, D., Oehmcke, S. & Andersen, I. G. Both sides of the story: Comparing student-level data on reading performance from administrative registers to application generated data from a reading app. *EPJ Data Sci.* **10**, 44 (2021).
8. Gissel, S. T. & Andersen, S. C. A cluster-randomized trial measuring the effects of a digital learning tool supporting decoding and reading for meaning in grade 2. *J. Comput. Assist. Learn.* **37**, 287–304 (2021).
9. Topping, K. Implementation fidelity in computerised assessment of book reading. *Comput. Educ.* **116**, 176–190 (2018).
10. Bulut, H. C., Bulut, O. & Arikan, S. Evaluating group differences in online reading comprehension: The impact of item properties. *International Journal of Testing* **23**, 10–33 (2023).
11. Keelor, J. L., Creaghead, N. A., Silbert, N. H., Breit, A. D. & Horowitz-Kraus, T. Impact of text-to-speech features on the reading comprehension of children with reading and language difficulties. *Annals of Dyslexia* **73**, 469–486 (2023).
12. Tüchler, A. F. & Cain, K. Effects of medium, text structure and participant characteristics on reading comprehension in young readers. *Journal of Research in Reading* **49**, e70029 (2026).
13. Beigman Klebanov, B. & Hoang, A. Toward stealth assessment of reading comprehension. *J. Res. Technol. Educ.* **58**, 73–92 (2026).
14. Paridon, J. van & Thompson, B. subs2vec: Word embeddings from subtitles in 55 languages. *Behavior Research Methods* **53**, 629–655 (2021).
15. Łuniewska, M. *et al.* Ratings of age of acquisition of 299 words across 25 languages: Is there a cross-linguistic order of words? *Behav. Res. Methods* **48**, 1154–1177 (2016).
16. Łuniewska, M. *et al.* Age of acquisition of 299 words in seven languages: American English, Czech, Gaelic, Lebanese Arabic, Malay, Persian and Western Armenian. *PLOS ONE* **14**, e0220611 (2019).
17. Chang, T. A., Arnett, C., Tu, Z. & Bergen, B. Goldfish: Monolingual language models for 350 languages. *Proceedings of the Language Resources and Evaluation Conference* 3750–3781 (2026) doi:10.63317/5ceec3hhv4d5.
18. Brysbaert, M., Warriner, A. B. & Kuperman, V. Concreteness ratings for 40 thousand generally known English word lemmas. *Behav. Res. Methods* **46**, 904–911 (2014).
19. Qi, P., Zhang, Y., Zhang, Y., Bolton, J. & Manning, C. D. Stanza: A Python natural language processing toolkit for many human languages. in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics: System Demonstrations* 101–108 (Association for Computational Linguistics, Online, 2020). doi:10.18653/v1/2020.acl-demos.14.
20. Chen, J. *et al.* M3-Embedding: Multi-linguality, multi-functionality, multi-granularity text embeddings through self-knowledge distillation. *Findings of the Association for Computational Linguistics ACL 2024* 2318–2335 (2024) doi:10.18653/v1/2024.findings-acl.137.
21. Buechel, S., Rücker, S. & Hahn, U. Learning and evaluating emotion lexicons for 91 languages. *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics* 1202–1217 (2020) doi:10.18653/v1/2020.acl-main.112.
22. Bernabeu, P. pilotr: Simulate experimental and behavioural data from a portable design specification. Zenodo https://doi.org/10.5281/zenodo.21266313 (2026).